

OpenPrivateer Player Manual

Public Alpha

OpenPrivateer is a modern reimplementation of *Wing Commander: Privateer*. It runs the game natively on modern systems while using the original game data.

OpenPrivateer does not include the original Origin/EA game assets. You need your own copy of *Privateer* and, if you want to play it, *Righteous Fire*.

This is a player manual. The reverse-engineering notes and file-format documentation in the source tree go much deeper.

1. Quick Start

1. Download and extract OpenPrivateer.
2. Start `openprivateer.exe` on Windows, or the corresponding executable on Linux or macOS.
3. The launcher opens. Select **Privateer** or **Righteous Fire** at the top.
4. Open **Game Data** and provide the original game files:
 - Privateer: `PRIV.TRE` and `INITCFG.PAK`
 - Righteous Fire: `RF.TRE` and `INITRF.PAK`
5. Use **Import to Data Folder...** to copy files from another folder, or **Add Folder...** to use files where they already are.
6. Click **PLAY**, then **NEW**.

The launcher shows the exact files it found and the folder each one comes from. You can also put the files next to the OpenPrivateer program; the launcher searches the Program Folder automatically.

The INIT files contain the original new-pilot starting state. If the matching INIT file is missing, **NEW** is disabled.

You do not need *Righteous Fire* to play the original campaign.

SDL3 on Linux and macOS

The Linux and macOS archives include the SDL3 library (`libSDL3.so*` or `libSDL3*.dylib`) next to the `openprivateer` program, and the program uses it automatically. To use your system's SDL3 instead (3.2.16 or newer), delete those `libSDL3` files from the program folder.

GOG installation

With the GOG release, `INITCFG.PAK` and `INITRF.PAK` are in the game installation folder. `PRIV.TRE` and `RF.TRE` are inside the `GAME.GOG` disc image and can be extracted with 7-Zip.

After extracting the TRE files, either collect the required files in one folder and import that folder, or add the folders containing them to the Game Data search path.

2. First Start and the Launcher

The launcher is OpenPrivateer's startup screen. It selects the game, locates the original game data and provides the main startup settings before the engine loads the game.

At the top, click the **Privateer** or **Righteous Fire** picture to choose the game. Your own `wallpaper/` artwork is used when present (section 19). On the left, the launcher shows the selected game's TRE and INIT file, their status and the exact path being used. **PLAY** stays disabled until the selected game's TRE file is found.

When a newer OpenPrivateer build is available, a line under the build number says **Update available** with the build's name and a **Download** button. Download opens the newest release in your web browser; nothing is downloaded or installed automatically (section 23).

The launcher sections are:

- **Experience:** choose Classic, Enhanced or Modern (section 4).
- **Game Data:** manage where OpenPrivateer finds the four original game files.
- **Music:** choose the music backend and an optional SoundFont (section 17).
- **Original Developer Switches:** set the original `-K`, `-B` and `-D` startup switches (section 24).
- **Files and Diagnostics:** open user-data folders, move the user-data location, reset settings, copy diagnostics and control update checks.

Untick **Show this screen at startup** to go straight to the game next time. Hold **Shift** while starting OpenPrivateer to bring the launcher back. The launcher is always shown when required game data is missing.

Game Data

OpenPrivateer searches the Game Data rows from top to bottom and uses the first copy of each required file it finds. The table shows all four files and marks the copy currently selected as **Used**.

The **Type** column describes each search location:

- **Custom:** a folder added by the player. Its files are used in place.
- **Data Folder:** the `data/` folder where the launcher stores imported or copied game files.
- **User Home:** the current user-data folder.
- **Program Folder:** the folder containing the OpenPrivateer program. In portable mode this is also the user-data folder.

Command-line and environment search locations are identified by their corresponding source instead.

Other table states help explain duplicate files. **differs** means a different copy exists further down the search path but is not used. **same** means the later copy is identical. **case!** reports a filename with the wrong letter case on systems where that matters.

Folders added as **Custom** can be moved up or down or removed. A path can also be entered directly and added with **Add**. **Rescan** checks the search locations again.

The three main file actions are different:

- **Add Folder...** adds a folder to the search path and leaves its files where they are.
- **Import to Data Folder...** selects a folder and copies recognized game files from it into the Data Folder.
- **Copy Found Files to Data Folder** copies the game files currently being used into the Data Folder.

When OpenPrivateer recognizes a likely existing installation, the launcher can also offer to use its files in place or copy the files it finds.

Where your files live

Settings (`openprivateer.ini`), savegames, logs and optional folders live together in OpenPrivateer's user-data folder.

By default this is:

System	Folder
Windows	<code>%APPDATA%\OpenPrivateer</code>
Linux	<code>~/.local/share/OpenPrivateer</code>
macOS	<code>~/Library/Application Support/OpenPrivateer</code>

A portable installation instead keeps user data in the Program Folder. An existing `openprivateer.ini` next to the program keeps older portable installations working. An empty `portable.txt` can also be placed there to request portable mode.

Open **Files and Diagnostics** to see the folder currently in use. The section has direct buttons for the user-data folder, savegames, logs, music, wallpaper and content.

Change Data Folder... selects another location for user data. When switching folders, OpenPrivateer can either use the destination as it is or copy the current settings, saves, content, music, wallpaper and copied game files there before switching. Nothing is deleted from the old location. Depending on the current mode, the launcher also offers **Move to User Folder...** or **Use User Folder...**

If the location was fixed with `--home` or `OPENPRIV_HOME` , it cannot be changed from the launcher for that run.

Reset Settings... renames `openprivateer.ini` to `openprivateer.ini.bak` and restarts OpenPrivateer. Savegames are not touched.

OpenPrivateer creates its optional folders and README files in the user-data folder automatically. The README files are rewritten on startup so their instructions stay in sync with the installed engine.

Folder	Purpose
<code>savegame/</code>	Original and OpenPrivateer savegames
<code>music/</code>	Optional replacement music
<code>wallpaper/</code>	Optional game-selection artwork
<code>content/</code>	Content overrides, translations, wingman names and the
	experimental <code>content/mods/universe/</code> map extension

Folder	Purpose
alpha/	Optional material from the August 1993 Privateer alpha
data/	Game files imported or copied by the launcher
logs/	Session logs

You can ignore all of these folders when starting out.

3. Privateer and Righteous Fire

OpenPrivateer supports both the original *Privateer* campaign and *Righteous Fire*.

Choose the game by clicking its picture at the top of the launcher. With the launcher turned off and both `PRIV.TRE` and `RF.TRE` found, a startup selector lets you choose instead. The selection is remembered. You can also change the game later from the settings/debug overlay. Changing games while playing relaunches OpenPrivateer because the two modes use different data archives.

Importing a Privateer pilot into Righteous Fire

The original Righteous Fire import flow is supported.

If Righteous Fire is selected and the `savegame/` folder contains Privateer `.SAV` files, starting **NEW** can ask whether you want to load an old Privateer ship.

The import carries over the pilot, ship, credits, reputation, kills, guild memberships and missions-to-date state according to the original import behavior. Privateer cargo, side missions and plot progress do not become Righteous Fire campaign state.

4. Experience Modes

OpenPrivateer has three experience settings: **Classic**, **Enhanced** and **Modern**. Enhanced and Modern currently use the same gameplay behavior; Modern is reserved for future changes.

Classic

Classic aims to reproduce the original gameplay, mechanics and feel. It is not a DOS emulator and does not deliberately reproduce every old hardware limitation or arithmetic accident. OpenPrivateer-only gameplay features such as multiple-ship ownership are inactive.

Enhanced and Modern

Enhanced and Modern keep the original game underneath but add OpenPrivateer conveniences and optional features.

Flight

- **Escape lands at a nearby base:** `Esc` in flight lands you at the nearest base, but only while you are inside that base's landing zone (5000 units, where "Entering automatic landing zone" appears). With no base that near it does nothing. In Classic, `Esc` does nothing in flight.
- **Cockpit hide:** `.` hides the cockpit background art while the HUD stays visible. It is unbound in Classic.
- **Rolled sprites keep their shape:** a planet stays round at any roll. Classic keeps the original oval distortion.
- **All input devices steer together:** keyboard, mouse and controller add up, so you can, for example, roll with the keypad keys while turning with the mouse or stick. In Classic only the device selected on the Options screen steers. Each device's own roll works the same in every mode: the mouse's right button and the original joystick's button 2 switch from turning to rolling, while a gamepad's right stick or a joystick axis assigned to Roll rolls while you turn.
- **Mouse flight is easier to control:** one even dead zone, the same sensitivity on both axes, and pushing the mouse up pitches the nose up. A second crosshair follows the pointer so you can see where you are steering, and **Mouse Sensitivity** applies. Classic keeps the original mouse flight: no visible pointer, and mouse up pitches the nose down.
- **Weapon state is remembered across a launch:** the gun on/off combination and last selected ordnance type are restored. Classic launches with all guns off.
- **Gun keys 1-4 address hardpoints:** each key flips one forward hardpoint. In Classic they flip a weapon-type group, so four identical guns all flip together on `1`.
- **Autopilot switches the tractor beam off.** In Classic it stays on.
- **Autonomous turrets:** a turret with no manual input slews toward the nearest hostile ship, leads it and fires. In Classic turrets are aimed by hand only.
 - Each turret has its own energy pool, so it cannot starve the forward guns. Classic uses one shared pool.
 - NPC turrets get a gunner too. In Classic they are fixed to the hull. **AI Turret Lead** in the debug overlay controls whether they lead.
- **Turret targeting uses the turret's aim:** `T` and auto-acquire in a turret view use the turret's own aim. Classic uses the ship's nose.
- **Other ships' missile and torpedo launches are audible** within 5000 units. In Classic they are silent.
- **Skipping a cutscene consumes the input:** the key or click that skips it does not also fire your guns. In Classic it also reaches the game, as in the original.
- **More cutscenes can be skipped:** the autopilot flyby and death cutscene can be skipped by any key or click. They cannot in Classic. The jump charge-up cannot be skipped in any mode.

Navigation

- **The Navigation Map opens while docked:** `N` opens it from any docked screen except the Mission Computer, where `N` types into the save name. Classic allows it only in flight.
- **Right-click zooms out:** Quadrant View to quadrant overview, then quadrant overview to sectors overview. This needs **Open Universe**.
- **`Shift` + arrow keys walk between quadrants** in Quadrant View (the plain arrows move the pointer, as on every pointer screen).
- **`Home` returns to the current system** in Quadrant View.

- **Shift +click a system** to show only its jump lines and list its destinations. A plain click shows every line again.
- **Hold Ctrl** to grey out systems your remaining jump fuel cannot reach, and **Ctrl +click** a system to see how many jumps away it is.
- **Hold Shift** to colour systems by faction.
- **Open Universe is on by default:** mod systems, jumps, bases and zones can become reachable. It is inactive in Classic. Built-in content currently adds Gemini in Humboldt, with the planet Gemini, the Heliopolis Array and Novaya Riga stations, two asteroid fields, random encounters and jumps to Troy, Junction and Aquila. **Open Universe** and **Built-in Content** in the debug overlay control this content. See section 20.

Bases and screens

- **Escape backs out one level** on base screens. It does nothing in Classic.
- **Mission Computer from Repair & Upgrade** is allowed. Classic blocks it.
- **Mission Computer REJECT button:** the alternate keypad can remove a mission from your list. Classic has only the original keypad.
- **No unreachable Mission Machine offers:** without a jump drive, the First Sector Mission Machine offers only missions inside the current system. Classic offers missions anywhere, as the original does. The guild boards are the same in every mode.
- **Side missions survive unrelated landings:** they are consumed only when they actually complete and pay. Classic drops every carried side mission on any landing.
- **Cargo Manifest last page:** it lists its items instead of the original's empty last page when the hold has exactly 8, 16 or 24 item types.
- **Bar brush-offs are local and temporary:** only that bartender or patron is affected, until you leave. In Classic one brush-off silences every bar for the rest of the career.
- **Entrance slides are quicker.** Classic uses the original 10 steps.
- **Clearer duplicate-purchase message:** buying a single-instance item you already have says **ALREADY OWNED**. Classic reuses the original **NO ROOM ON SHIP** message.
- **Higher ordnance capacity:** missile and torpedo capacity is 20 per matching launcher. Classic uses the original 10 per launcher.
- **Faster hardpoint-purchase cancel:** right-click cancels the Ship Dealer hardpoint picker. Classic keeps only the on-screen **Never mind** region.

Enhanced and Modern also permit multiple ships and hired wingmen when their settings are enabled. Use **Classic** for direct comparison with DOS Privateer, and **Enhanced** or **Modern** for OpenPrivateer's extra features.

The Experience setting can be changed while playing. In multiplayer it is fixed by the server.

5. Settings and Debug Overlay

During the Alpha, some settings still live in the debug/settings overlay rather than a finished front end.

Press:

~

This is normally the key to the left of **1** on an English keyboard. Its label depends on your keyboard layout.

The overlay contains settings for areas such as Experience mode, Privateer / Righteous Fire mode, audio, SoundFont, rendering, subtitles, Mission Computer style, controllers, multiple ships, hired wingmen and update checking.

Some settings are disabled in Classic because they would change original gameplay.

6. Controls

OpenPrivateer supports keyboard, mouse, gamepads, joysticks and HOTAS-style controllers.

Open **Controls...** from the settings interface to inspect or remap controls. Controller bindings are stored per device. The same screen can also be reached by pressing **K** on the Options screen, opened in flight or on a base with **Alt+O**.

Assigning a controller

Selecting a controller and assigning it are separate steps:

1. Open **Controls...**, then **Remap Game Controls**.
2. Choose the physical controller from the controller drop-down at the top.
3. Click **Assign Controller**.

The drop-down only selects which connected device the Controller column shows and edits. **Assign Controller** makes that device active in OpenPrivateer. With several controllers connected, check the drop-down before changing a binding.

Assigning buttons, hats and axes

After assigning the controller:

1. Find the game action you want to change.
2. Click one of its slots in the **Controller** column.
3. Press the desired button, move the hat, or move the desired axis.

For example, to assign a joystick X axis to Yaw: select and assign the joystick, click the Controller slot on the **Yaw** axis row, then move the joystick's X axis.

Right-click a binding slot to clear it. **Esc** cancels capture without changing the slot. If the input you press is already used by another action that works at the same time (for example two flight actions), it is moved to the new action and the status line says where it was removed from. Actions that never work at the same time can share an input: a key can mean one thing in flight and another on a base screen or in the Mission Computer. Global keys such as fullscreen and quit work everywhere, so they share with nothing. **Clear Controls** clears the selected controller's bindings. **Remove** makes it inactive; it stays in the list and can be assigned again later.

OpenPrivateer can bind controller buttons, hats, axis directions and full analog axes. On Pitch, Yaw, Roll and Throttle, the round button next to the controller axis reverses it (filled = reversed).

Calibrating an axis or throttle

A full controller axis must be assigned before **Calibrate** becomes available. The complete order is therefore:

1. Select the controller in the drop-down.
2. Click **Assign Controller**.
3. Click the Controller slot on Pitch, Yaw, Roll or Throttle.
4. Move the physical axis to assign it.
5. Click **Calibrate** on that row.

The dialog shows the live raw value. Confirm with **Enter**, **Confirm**, or a tap of any button on that controller. To cancel, hold any button on that controller for about 1.5 seconds (a bar fills while you hold it), or press **Esc** or **Cancel**. Cancelling keeps the previous calibration.

- **Pitch, Yaw, Roll:** move through the full travel in both directions, let the control rest at centre, then confirm. OpenPrivateer records minimum, centre and maximum.
- **Throttle:** confirm three points: **zero**, **full military power** and the point where **afterburner** should begin. Choose **No afterburner zone** if the lever has no afterburner detent. The order establishes the lever direction, so a calibrated throttle normally does not need the reverse control.

Throttle axis modes

The Throttle row has a **Mode** button:

- **Auto** (default): Absolute for a joystick/throttle axis, Relative for a gamepad stick.
- **Absolute:** lever position directly sets speed. Moving past full military power into a calibrated afterburner zone engages afterburner.
- **Relative:** deflection from centre raises or lowers speed, like the original joystick throttle. On a calibrated throttle the calibrated **zero** point is the centre: moving towards full military power raises speed, moving the other way lowers it.

In Absolute mode the lever only takes over after you move it. Starting a flight does not immediately force the ship to the lever's physical position. Keyboard and mouse throttle controls still work; moving the lever again gives it control.

Calibration

OpenPrivateer also supports general joystick calibration. A raw joystick needs at least two axes and one button to qualify as a flight controller. SDL-recognized gamepads are accepted directly.

If several devices are attached, the calibration screen calibrates the **Primary Flight Controller** (see Controller settings). With **Automatic**, devices that cannot actually fly the ship are skipped.

The first time the calibration screen appears on its own (Options set to Joystick and the controller never calibrated), the Controls settings open on top of it. Pick your Primary Flight Controller there and press

Done: the calibration screen is waiting underneath and calibrates the controller you chose. While the settings are open, the calibration screen ignores the controller.

On the calibration screen, tap a button to confirm each step. Hold a button for about 1.5 seconds to cancel; a bar fills while you hold it. `Esc` also cancels. Cancelling works at any step in every experience mode and keeps the previous calibration.

Controller settings

- **Dead Zone:** in Enhanced and Modern a round dead zone around the stick centre, adjustable from 3 to 30 percent (default 15), so a diagonal push is not cut off on one axis. Classic uses the original 30 percent dead zone on each axis separately.
- **Response Curve:** Linear or Smooth in Enhanced and Modern. Smooth gives finer control near the centre and full rate at full throw.
- **Mouse Sensitivity:** for mouse flight in Enhanced and Modern.
- **Input Backend** on Windows: WinMM or SDL DirectInput.
- **Primary Flight Controller:** which controller is “the joystick”. It flies with the original stick scheme (trigger, button 2 roll, double-tap afterburner), moves the menu pointer, is the one the whole-joystick calibration screen calibrates, and in Classic is the Joystick input device. **Automatic** (the default) picks the first connected controller that can fly. The list updates when you plug a controller in or out; two identical controllers show as “(1)” and “(2)”. The choice applies immediately. It does not change any button or axis binding: those stay with the controller they were assigned on, set with **Assign Controller**. If the chosen controller is not connected, the game uses Automatic’s choice for now, keeps your choice for next time, and the page says which controller it is using. A newly chosen controller that has never been calibrated is not calibrated automatically; use the calibration screen for it.

Default gamepad layout

Input	Default action
Left stick Y	Pitch
Left stick X	Yaw
Right stick X	Roll
Right stick Y	Throttle
Right trigger	Fire guns
Right shoulder	Fire ordnance
Left trigger	Afterburner
Left shoulder	Cycle target
South button	Target lock
East button	Cycle ordnance
West button	Cycle guns
North button	Toggle ITTS

Input	Default action
Start	Pause
Back	Navigation Map
D-pad up	Front view
D-pad down	Rear view
D-pad left	Left view
D-pad right	Right view

These are normal remappable bindings. A joystick or HOTAS that SDL does not recognize as a gamepad gets the same four views on its first hat switch. If you have put something else on a D-pad direction, your binding stays and that view default is left off. The views are flight-only; on the menu screens the D-pad (and the first hat) moves the pointer in Enhanced and Modern.

Joystick pointer

A calibrated joystick can move the pointer on base and menu screens, matching a feature of the original game. The joystick trigger acts as the click. SDL gamepads use the A/South button.

Keyboard pointer

As in the original game, the arrow keys move the pointer on every screen that shows it except the Mission Computer, and **Enter** clicks. The pointer moves in small steps and speeds up while you hold the key. Holding **Enter** while moving drags. This works in every experience mode.

In Enhanced and Modern the keys are remappable (the **Pointer** group on the remap page), and these also work:

Input	Action
Numeric keypad 8, 2, 4, 6	Move the pointer
Numeric keypad 5	Click
Numeric keypad 0	Right click
D-pad / first hat	Move the pointer

The pointer also works at a conversation's Accept/Reject prompt there. On the Navigation Map the plain arrows move the pointer, so the Enhanced/Modern quadrant walk uses **Shift** + arrows.

7. New Game, Load and Save

The title screen has **NEW**, **LOAD**, **OPTIONS** and **QUIT**.

NEW

NEW starts from the original data in **INITCFG.PAK** or **INITRF.PAK**. If the matching file is missing, NEW is greyed out.

LOAD

Privateer saves use `.SAV`. Righteous Fire saves use `.PRS`.

Put existing DOS saves in the `savegame/` folder of the user-data folder (the launcher's **Savegames** button opens it):

```
savegame/
```

OpenPrivateer reads the original save format.

SAVE

Use the Mission Computer and choose **SAVE**.

Saving under a typed name never overwrites a file. Matching the original game, the name always gets `_NEW` appended (then `_NEW2`, `_NEW3`, ...) rather than reusing the plain name. Typing `PILOT` creates `PILOT_NEW.SAV`, not `PILOT.SAV`; saving under that same name again creates `PILOT_NEW2.SAV`, and so on.

Some OpenPrivateer-only state cannot fit inside the original DOS save format. That state is stored in a matching `.openpriv.json` sidecar while the `.SAV` or `.PRS` remains in the original format.

If you use parked ships or other OpenPrivateer-only persistent features, keep the sidecar together with its save.

8. Mission Computer

The Quine 4000 handles saves and pilot information.

Its important pages are:

- **MISSIONS**: active plot mission and accepted side missions
- **FIN**: cash, kills and faction standing
- **MAN**: cargo manifest
- **SAVE / LOAD**: save browser

A pilot can carry up to three side missions. The active plot mission is separate and does not consume one of those slots.

Classic uses the original Mission Computer presentation. Enhanced and Modern can use the alternate OpenPrivateer presentation through the **Quine Model** setting.

9. Bases

Bases use the original game's scenes, people and layout.

Move the pointer over doors, people and terminals to see available actions, then click normally.

Depending on the base you may find the Launch Pad, Main Concourse, Bar, Commodity Exchange, Ship Dealer, Repair & Upgrade, Mission Computer, Merchants' Guild or Mercenaries' Guild.

Not every base contains every service.

10. Commodity Exchange

The Commodity Exchange buys and sells cargo. Left-click "Select Items" to buy or sell one unit. Right-click it to buy as many as you can afford and carry, or to sell everything you hold of that commodity, in every experience mode.

Prices and availability depend on the base economy. Cargo consumes hold space. Mission cargo follows the game's normal cargo rules.

The secret compartment, once obtained through the story, has separate hidden capacity.

11. Ship Dealer and Repair & Upgrade

With normal single-ship play, buying another hull follows the original trade-in model. The current hull and sellable equipment contribute to the trade-in value.

Repair & Upgrade handles ship equipment.

Buy mode lists only what your hull can take, as in the original: each hull has its own shield and engine ceiling and its own turrets, and the Centurion cannot fit the Cargo Expansion. Sell and Repair list whatever the ship carries.

In Classic, entering Repair & Upgrade always starts in **Buy** mode on the first item, matching the original. Enhanced and Modern remember the previous mode and item where possible.

A damaged scanner can be repaired in the Software Booth's Repair Mode. The cost depends on the scanner and how badly it is damaged, as in the original game.

12. Multiple Ships and Fleet Selector

Multiple ships are an OpenPrivateer feature.

To use them:

1. Switch to Enhanced or Modern.
2. Enable **Allow Multiple Ships**.
3. Buy another hull.

Instead of trading in the old ship, OpenPrivateer parks it at the current base and makes the new hull active.

On a Launch Pad press:

S

to open the Fleet Selector.

The active ship is marked as current. Parked ships can be selected or sold. Remote selection of a ship parked at another base is supported.

Multiple ships are not active in Classic.

13. Hired Wingmen

Hired wingmen are an OpenPrivateer feature built on top of multiple-ship ownership.

Enable **Allow Multiple Ships** and **Allow Hired Wingmen** in Enhanced or Modern mode.

A hired pilot flies one of your actual owned parked ships. If the wingman survives, the ship returns with its live equipment state. If it is destroyed, the ship is lost.

Pilots have persistent callsigns and skill. Confirmed kills can improve their skill. Launching with a wingman costs a sortie fee.

A ship assigned to a hired pilot cannot simply be selected or sold as an ordinary parked ship.

Custom callsigns

The callsign pool is stored in:

```
content/wingmen/callsigns.json
```

OpenPrivateer creates the default file if it is missing and does not overwrite an edited version.

The file is a JSON array of non-empty strings. It is read as JSON5, so `//` and `/* */` comments and a comma after the last name are allowed:

```
[
  "Rook",
  "Nova", // my favourite
]
```

If the file has an error, the built-in names are used and the log says where (line and column). Existing pilots keep their callsigns. Changes affect later hires.

Plain ASCII names are safest with the original game fonts.

14. Flight

Flight follows the original game's own flight model.

Flight includes throttle, afterburner, guns, ordnance, targeting, ITTS, radar, scanners, navigation, autopilot, jumps, docking, tractor beam, turrets, equipment damage and the Repair Droid.

To jump, fly within range of a jump point and press **J**. As in the original, **J** does nothing at all when no jump is possible there (no message); only an empty jump-drive tank shows "No fuel" on the Destination page.

Shields

S steps the shield generator's power level down and **Shift + S** steps it back up, one level at a time, in every experience mode, as in the original. A lower level holds less charge per facing but draws less energy; level 0 switches the shields off. A smaller generator has fewer levels.

Scanner

Targeting requires a working scanner.

With no scanner fitted, the radar is absent and scanner-dependent targeting functions are unavailable. There is no hidden free scanner.

Tractor beam

Select a cargo canister or an ejected pilot with **T** first; it does not need to be locked. With nothing selected, or with a ship, base or planet selected, **Enter** does nothing. A tractor beam in a turret is switched on and off with **Enter** from that turret's view (**F5** / **F6**). A tractor beam on one of the ship's own launcher hardpoints is selected with **W** from a forward view - it comes after the missiles and torpedoes and the Weapons display reads **W: Tractor Beam** - and then switched on and off with **Enter**. A beam keeps pulling the object it was switched on against, even if you change views or select something else. It switches itself off without a message when that object is gone, more than 10000 units away, does not fit in the hold, or the ship runs out of energy. A beam on a launcher hardpoint is drawn from that hardpoint, below the nose.

As in the original, several tractor beams can run at once and each one is drawn. Each beam that pulls an object in adds it to the cargo hold, so two beams on one object give you two of it.

Autopilot

Autopilot follows the original instant-travel behavior. It can drop out early when the route enters another encounter zone, including asteroid or navigation encounter areas.

Pause

Press **P** in flight.

15. Navigation Map

The Navigation Map uses the original sector and quadrant data.

It shows systems, nav points, bases, jump points and mission objectives. As in the original, only jump points are blue circles; bases are green squares, and pirate bases and empty nav points are green circles. Story-locked systems and routes remain hidden until the game makes them available. Once a story jump point is shown (the Kilrathi systems beyond Rygannon), it can be flown through like any other jump point.

Active mission targets are highlighted.

The optional **Open Universe** setting exposes additional universe content and is not part of Classic behavior.

16. Guilds and Random Missions

The Merchants' Guild, Mercenaries' Guild and Mission Computer provide the original non-plot mission types:

- Cargo
- Patrol
- Scout
- Defend
- Attack
- Bounty

Guild fees are:

Guild	Fee
Merchants' Guild	1,000 credits
Mercenaries' Guild	5,000 credits

Both guild computers show the original **Scanning ... records ... Please stand by...** stage while generating a fresh batch of offers.

17. Music and Sound

OpenPrivateer supports several music backends, including OPL2, MIDI with an external SoundFont, OS MIDI and CD-style replacement music. Availability depends on the platform.

SoundFont

Select a `.sf2` file in the launcher's **Music** section or from the SoundFont setting when using the corresponding MIDI backend. The path is stored in `openprivateer.ini`. Without one, OpenPrivateer looks for `openprivateer.sf2` in the user-data folder, then next to the program.

Replacement music

The `music/` folder accepts optional replacement music. WAV, MP3 and FLAC are supported.

Use the generated `music/README.txt` for the current slot and filename rules. That README is kept in sync with the installed build.

18. Graphics

OpenPrivateer renders the original game at its native logical resolution and scales it for modern displays.

Available settings include CRT filtering, image adjustments, scaling, texture filtering, pixel scaling and splash font selection.

Classic keeps the original's look, including its quirks. Enhanced and Modern apply some presentation improvements, such as keeping round sprites round while the ship rolls.

19. Wallpaper

The `wallpaper/` folder accepts custom artwork for the Privateer / Righteous Fire selector.

Use filenames beginning with:

```
01_
```

for Privateer and:

```
02_
```

for Righteous Fire.

PNG, JPG and BMP are supported. See `wallpaper/README.txt` for the current details.

20. Content Overrides and Translations

OpenPrivateer can load replacement files from the filesystem without changing the original TRE archives.

The basic layout is:

```
content/

  priv/

  rf/

  i18n/

  mods/universe/
```

`content/priv/` and `content/rf/` mirror paths inside the corresponding original archive.

`content/mods/universe/systems.json` adds systems and jump lines to the Open Universe. The file is JSON5, so comments, trailing commas, unquoted field names, single-quoted strings and hex numbers are allowed, but each field may appear only once per object. Invalid files are rejected as a whole and the log names the field, or the line and column of a typo. Delete a file to turn it off.

`content/mods/universe/factions.json` adds factions with their own map color and name (shown as "Faction" in a selected system's info panel). A system in `systems.json` can name one in its optional `faction` field.

`content/mods/universe/navpoints.json` adds bases, planets, stations, plain nav points (empty space, an asteroid field) and jump points. Added bases can be flown to and landed at. A mod base can reuse the interior of a similar original base while keeping its own location identity. A nav point can be made a no-weapons-fire zone.

`content/mods/universe/zones.json` adds asteroid fields to systems. A zone can use a position or a nav point to sit on, a size and a density ("sparse" , "medium" , "dense" or 1 to 7 rocks).

`content/mods/universe/encounters.json` adds random encounters: for a nav point or zone, a list of possible groups with their chance, number of ships, faction, ship type, skill and morale. Each time you enter the system one group per nav point is picked, and its ships appear when you come near that point, as in the original systems.

The generated `content/mods/universe/README.txt` contains the current formats and built-in examples.

OpenPrivateer also ships built-in Open Universe content. In Enhanced and Modern with Open Universe enabled, **Gemini** is a flyable system in Humboldt, the metropolis of the sector. Its nav points:

Nav	What is there
1	Gemini , the metropolis planet (landing base)
2	Heliopolis Array , a solar array station (no weapons fire)
3	Novaya Riga , a run-down refinery station
4, 5	Asteroid fields
6, 7	Empty space
8, 9, 10	Jump points to Troy, Junction and Aquila

Return jumps connect those systems back to Gemini: Troy at Nav 9, Junction at Nav 10 and Aquila at Nav 1. You can fly through Gemini with autopilot and leave through any of its three jump points. On the Navigation Map, QUAD and `Home` show the quadrant of the system you are in (Humboldt for Gemini, Petrov for Aquila). Selecting one of Gemini's nav points shows "Jump to " for a jump point, the station's name and type for a station, and "Asteroid Field" or "Empty Space" for the plain nav points. Gemini's sky shows no sun or planets yet.

Gemini has traffic like an original system. Near each nav point you may meet militia patrols in Gladius fighters, merchants in Tarsus and Drayman transports, or pirates in Talons and in Demon fighters; around the planet a militia Paradigm sometimes patrols. Pirates are most common around Novaya Riga and the asteroid fields, the militia around the planet and the stations. Which group you meet at a nav point is decided each time you enter the system, and its ships appear when you come near the point. Autopilot stops when its route crosses another nav point's area, as in an original system.

Weapons cannot be fired near the **Heliopolis Array**: within about 12000 meters of it your guns, turrets and missiles stay silent and the message line reads "No weapons fire zone". Other ships are not bound by it. The tractor beam still works.

A visit to a built-in mod base uses the interior of a similar original base, but the mod base keeps its own name and location and only its configured services are available. Its Mission Computer and guilds have no missions yet, and missions cannot be delivered there. You launch back into Gemini. Saving while docked at a mod base stores the extra location in the matching `.openpriv.json` sidecar while the original save stays in the original format. If you turn the built-in content or Open Universe off before loading, you load at the similar original base instead.

The asteroid fields at Nav 4 and Nav 5 work like original asteroid fields: rocks can hit the ship, and autopilot stops when it reaches a field.

A `navpoints.json`, `zones.json` or `encounters.json` that reuses a built-in key with the same `mod` and `key`, adds a second jump between the same systems or gives a built-in nav point or zone a second encounter table, is not loaded.

To disable the built-in content, uncheck **Built-in Content** under **Open Universe** in the debug overlay, or set `UniverseBuiltinContent=0` in `openprivateer.ini`, then restart.

Translations live under `content/i18n/` with names such as:

```
openprivateer_de.i18n
```

The translation format uses `key=value`, `#` comments and `\n` for embedded line breaks.

The original fonts have limited character coverage. Plain ASCII or transliteration is safest unless a suitable custom font is used.

21. Optional 1993 Alpha Content

The `alpha/` folder is for optional material from the August 1993 Privateer alpha.

It is not distributed with OpenPrivateer and is not required to play.

See `alpha/README.txt` for the exact files expected by your build.

22. Multiplayer

OpenPrivateer includes dedicated client/server multiplayer. It is still an Alpha feature.

Server

Basic example:

```
openprivateer_server --tre /path/to/PRIV.TRE
```

The default UDP port is `24601`.

Useful server options include `--port`, `--max-players`, `--verbose` and `--experience-mode`.

Client

Basic example:

```
openprivateer --connect HOST:24601 /path/to/PRIV.TRE
```

The server can reject clients when important session properties differ, including build, Privateer vs. Righteous Fire data or Experience mode.

Multiplayer covers the shared flight/combat environment. The normal single-player campaign progression is not the multiplayer campaign model.

23. Update Check

Release builds can check for a newer OpenPrivateer build at most once every 24 hours.

Nothing is downloaded or installed automatically.

When a newer build exists, the title screen shows:

```
New build available
```

```
Click to download
```

Left-click opens the download page. Right-click dismisses that particular build notice.

The launcher shows the same news as an **Update available** line with a **Download** button. On Windows and Linux, Download opens your browser on the matching archive of the newest build; on other systems it opens the release page, which lists every archive. Unpack the archive yourself to update.

The check can be disabled in settings or for one launch with:

```
--no-update-check
```

24. Original Developer Switches

OpenPrivateer supports several switches from the original game:

```
-K
```

Skip the intro.

```
-B<n>
```

Start at original base ID `n`.

```
-D<n>
```

Set starting credits to `n`.

Example:

```
openprivateer.exe -K -B39 -D100000
```

These are mainly useful for testing and development.

The launcher's **Original Developer Switches** section sets the same three as checkboxes that are remembered and apply on every start. Switches given on the command line win. The `-B` and `-D` number fields can only be edited while their box is ticked.

Other command-line options

```
openprivateer --help
```

lists every option. The ones for files and the launcher:

Option	Effect
<code>--home DIR</code>	Use <code>DIR</code> as the user-data folder for this run
<code>--portable</code>	Use the program's folder as the user-data folder
<code>--data DIR</code>	Look for the game files in <code>DIR</code> first
<code>--launcher</code>	Show the launcher
<code>--no-launcher</code>	Never show it; exit with an error if files are missing

`OPENPRIV_HOME` and `OPENPRIV_DATA` (a list separated like `PATH`) are the environment-variable versions of `--home` and `--data`.

25. Debug Tools

The Alpha contains debugging tools used during reconstruction and testing.

Open the settings/debug overlay with:

~

Toggle the base click-region overlay with:

R

The region overlay draws clickable areas in green and is useful for screenshots when reporting a base UI problem.

26. Troubleshooting

NEW is greyed out

The matching INIT file is missing. Privateer needs `INITCFG.PAK` ; Righteous Fire needs `INITRF.PAK` . Copy it in with the launcher (hold Shift while starting if it is turned off), or put it next to its `.TRE` file.

The game data cannot be found

The launcher lists every folder it searched and which files it found. Use **Import to Data Folder...** to copy files in, or **Add Folder...** to use them where they are. On Linux, a file named `priv.tre` is reported as "wrong letter case": rename it to `PRIV.TRE` .

With the GOG release, `PRIV.TRE` and `RF.TRE` are inside `GAME.GOG` and can be extracted with 7-Zip. The two INIT PAK files are in the game installation folder.

Where are my settings and saves?

Open the launcher's **Files and Diagnostics** section. It shows the user-data folder currently in use and provides buttons for the folder, savegames, logs, music, wallpaper and content.

An old save does not appear

Privateer saves must be `.SAV` , Righteous Fire saves `.PRS` . Put them in the `savegame/` folder of the user-data folder.

"libSDL3.so.0: cannot open shared object file"

On Linux this means the program found neither the bundled SDL3 nor a system SDL3. Usually the archive was not extracted completely: extract it again so the `libSDL3.so*` files sit next to `openprivateer` , or install SDL3 3.2.16 or newer from your distribution.

The controller is detected but does not fly the ship

Open **Controls...**, check the device and calibration, then verify the axis bindings. A raw joystick needs at least two axes and one button.

MIDI is silent

Check the selected music backend. If using SoundFont MIDI, make sure a valid `.sf2` file is selected.

Something behaves differently from DOS Privateer

Check the Experience mode first. Use **Classic** for direct comparisons with the original.

If the difference remains in Classic, report it.

Windows Defender reports the executable

New unsigned software can occasionally trigger heuristic antivirus detections. Do not blindly whitelist a file.

Make sure the binary came from the official OpenPrivateer release, then update Windows Security's security intelligence and scan it again. A signature update can clear such a detection.

If a current official build is still detected, report the build ID and exact detection name.

27. Reporting Bugs

A useful Alpha bug report includes:

- OpenPrivateer build ID
- Privateer or Righteous Fire
- Classic, Enhanced or Modern
- what you expected
- what happened
- exact reproduction steps
- whether it happens every time
- the savegame when relevant
- a screenshot or video for visual problems
- for startup or file problems, the text from the launcher's **Copy Diagnostics** button

Print the build ID with:

```
openprivateer --version
```

For base-screen problems, a screenshot with the `R` region overlay can help.

You do not need to decide whether something is an OpenPrivateer bug or an original Privateer quirk before reporting it. Describe what happened and it can be checked against the original.

28. What Alpha Means

OpenPrivateer is playable, but it is still an Alpha.

The normal single-player Privateer path is implemented, and Righteous Fire support is under active reconstruction and testing. Obscure cases are still being found as more people play the game.

Expect bugs, unfinished polish and occasional behavior changes as the original game becomes better understood.

Save before doing anything adventurous.

29. Help and Downloads

Project website:

<https://openprivateer.org/>

Downloads:

<https://openprivateer.org/#downloads>

Discord:

<https://discord.gg/5H3vW2STqW>

When asking for help, include the build ID and whether you are playing Privateer or Righteous Fire.

30. Legal Note

OpenPrivateer is a reimplementations of the game engine.

It does not ship the copyrighted game data from *Wing Commander: Privateer* or *Righteous Fire*. You must provide the original game files yourself.

Wing Commander, *Privateer*, *Righteous Fire* and related names and assets belong to their respective rights holders.

31. One-Minute Checklist

For Privateer:

- ☐ Extract OpenPrivateer
- ☐ Start OpenPrivateer
- ☐ Launcher: copy in PRIV.TRE and INITCFG.PAK
- ☐ Leave Experience on Classic for original behavior
- ☐ Click PLAY, then NEW

For Righteous Fire:

- ☐ Launcher: copy in RF.TRE and INITRF.PAK
- ☐ Select Righteous Fire

For an old pilot:

- ☐ Launcher: Files and Diagnostics, Savegames
- ☐ Copy .SAV or .PRS there
- ☐ Click LOAD

For OpenPrivateer's extra features:

- ☐ Switch to Enhanced or Modern
- ☐ Open the settings overlay with ~
- ☐ Enable the features you want

Have fun.